Extremal Degree Moments in Directed Graphs

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Abstract

We study sums of convex functions of out-degrees in digraphs with a forbidden subdigraph. For every tournament on five vertices, the sum of squared out-degrees is at most $9n^3/16$. For sufficiently large orders, the unique extremal digraph is the balanced complete bidirected four-partite graph. We also prove a quantitative stability theorem for all twelve five-vertex tournaments. A single rational flag certificate controls a larger class defined by a local five-vertex condition. For the bidirected join of a directed triangle and any number of directed edges, we determine every nondecreasing convex out-degree extremum at every order, classify all square-sum extremizers, and prove their qualitative edit-distance stability. These results extend the known directed-triangle case to an infinite family whose square-sum extrema exceed a proposed bound based on the ordinary Turán number.

1 Introduction

All digraphs in this paper are finite and loopless, with at most one arc in each direction between two vertices. Opposite arcs are allowed. A pair carrying both arcs is a *digon*. A digraph is *semicomplete* if every pair of distinct vertices is adjacent. We write $d_D^+(v)$ for the out-degree of v and

$$Q(D) = \sum_{v \in V(D)} d_D^+(v)^2.$$

A copy of a forbidden digraph is a subdigraph, not necessarily induced. For a nondecreasing convex function $\varphi : [0, n-1] \to \mathbb{R}$, define

$$\mathrm{ex}_\varphi^+(n, F) = \max\Big\{ \sum_{v \in V(D)} \varphi(d_D^+(v)) : |V(D)| = n,\ D \text{ is } F\text{-free} \Big\}.$$

The notation $\mathrm{ex}_2^+(n, F)$ denotes the square case. We write $\mathrm{ex}(n, F)$ for the corresponding maximum number of arcs.

Lin, Wang, Zhou, and Zhou [5] studied degree-square bounds in connection with uniformly dense hypergraphs and asked for the extremum under a general forbidden digraph. Ai, Chen, Chen, Yan, and Zhao [1] developed a prefix-majorization method for convex out-degree functionals and asked which self-converse tournaments admit an exact evaluation. A tournament is self-converse if reversing all its arcs gives an isomorphic tournament. Iľkovič [4] treated four-vertex tournaments, and Han and Chen [3] determined the square-sum extremum for every transitive tournament and for the regular tournament on five vertices.

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Let $\overleftrightarrow{T}_r(n)$ be the complete bidirected r-partite digraph with part sizes differing by at most one; parts are independent, and each cross pair is a digon. Put $B_4(n) = Q(\overleftrightarrow{T}_4(n))$. If $n = 4a + b$, where $0 \le b < 4$, then

$$B_4(n) = b(a+1)(3a+b-1)^2 + (4-b)a(3a+b)^2.$$

Our first result treats all five-vertex tournaments, including those that are not self-converse. Distance between two digraphs on the same vertices means the number of individual arc additions and deletions needed to transform one into the other.

Theorem 1.1. *Let F be any tournament on five vertices. Every n-vertex F-free digraph satisfies $Q(D) \le 9n^3/16$. For $n \ge 1$, equality holds exactly when $4 \mid n$ and $D \cong \overleftrightarrow{T}_4(n)$. For every $n \ge 526$,*

$$\mathrm{ex}_2^+(n, F) = B_4(n),$$

and $\overleftrightarrow{T}_4(n)$ is the unique extremal digraph up to isomorphism. There is an absolute constant C such that, whenever $\varepsilon \ge 0$ and $Q(D) \ge (9/16 - \varepsilon)n^3$, the digraph D is at distance at most $C\varepsilon^{1/4}n^2 + 3n$ from a copy of $\overleftrightarrow{T}_4(n)$.

The exact values for the transitive and regular five-vertex tournaments are already contained in [3]. The theorem extends the common extremal description to every five-vertex tournament and proves a uniform stability statement, addressing the question in [3, Section 6] in this range. The order threshold is not intended to be optimal.

The upper bound and stability follow from a stronger local statement. Let $\mathcal{C}$ consist of the digraphs for which every induced five-vertex subdigraph fails to contain at least one of the twelve five-vertex tournaments. The missing tournament may depend on the five-set. Every digraph avoiding a fixed five-vertex tournament belongs to $\mathcal{C}$. A rational certificate on all 9558 allowed five-vertex types proves the sharp bound and gives a positive gap on each type that is not complete bidirected multipartite. We turn this local gap into an explicit edit estimate. The finite exact result then follows from a cloning argument and the same gap.

Our second result concerns forbidden digraphs with digons. For vertex-disjoint digraphs A, B, their *bidirected join* $A \oplus B$ adds both arcs between every vertex of A and every vertex of B. Let C_3 be a directed triangle, let T_2 be a single directed edge, and set

$$F_q = C_3 \oplus \underbrace{T_2 \oplus \cdots \oplus T_2}_{q-1 \text{ copies}}, \qquad q \ge 1.$$

An *ordered digon chain* L_s consists of successive two-vertex digon blocks, followed by one singleton if s is odd, with all arcs directed from each earlier block to each later block. We allow L_0 to be empty. Write $t_r(n)$ for the number of edges in the balanced complete r-partite undirected graph.

Theorem 1.2. *For every $q, n \ge 1$,*

$$\mathrm{ex}(n, F_q) = 2t_{2q}(n). \tag{1}$$

For every nondecreasing convex $\varphi : [0, n-1] \to \mathbb{R}$,

$$\mathrm{ex}_\varphi^+(n, F_q) = \sum_{j=0}^{n-1} \varphi\left(n - 1 - 2\left\lfloor \frac{j}{2q} \right\rfloor\right). \tag{2}$$

Write $n = 2qa + b$, where $0 \le b < 2q$. The square-sum extremizers are precisely

$$L_{2a+s_1} \oplus \cdots \oplus L_{2a+s_q}, \qquad s_i \in \{0,1,2\}, \quad \sum_{i=1}^{q} s_i = b, \tag{3}$$

up to isomorphism. The same classification holds for strictly convex, nondecreasing functions. The in-degree extremum has the same value, and its square-sum extremizers are the reversals of the digraphs in (3).

For $q = 1$, the value and chain structure are known from [1]. We also use their general prefix-majorization principle for convex degree sums. The work needed for $q \geq 2$ is to construct a graph attaining every prefix bound for F_q, recover all graphs with equality, and control their near-extremizers. The equality classification can contain more than one graph. For example, $L_4 \oplus L_2$ and $L_3 \oplus L_3$ are both extremal when $q = 2$ and $n = 6$. Theorem 4.2 shows that, for each fixed $q \geq 2$, every square-sum near-extremizer is close in edit distance to a balanced join of q ordered digon chains. The corresponding $q = 1$ stability results are known from [1, 4].

In the discussion preceding Question 5.2 of [5, Section 5], the authors propose that the transitive-tournament bound $((r-2)/(r-1))^2 n^3$ should extend to every r-vertex forbidden digraph whose underlying graph is complete and whose ordinary extremum is $2t_{r-1}(n)$ for every n. Theorem 1.2 gives such forbidden digraphs of arbitrarily large odd order, but

$$\lim_{n\to\infty} \frac{\mathrm{ex}_2^+(n, F_q)}{n^3} = 1 - \frac{1}{q} + \frac{1}{3q^2} = \left(1 - \frac{1}{2q}\right)^2 + \frac{1}{12q^2}. \tag{4}$$

The known directed-triangle result already contradicts this proposal when $q = 1$. Formula (4) gives counterexamples of every odd forbidden order at least five as well. Question 5.2 itself asks for the maximum of the out-degree and in-degree square sums under a prescribed forbidden digraph. Theorem 1.2 answers that question for the family F_q, since its members are self-converse and the two extrema agree.

Section 2 proves the local square-sum bound and the finite extremal result. Section 3 derives stability for the original digraph, without assuming a multipartite structure. Section 4 proves the convex moment formula, classifies its extremizers, and establishes stability for the chain joins. The computer-assisted part is confined to one explicitly described finite certificate; all subsequent deductions are analytic or combinatorial.

2 The five-vertex bound

We first specify the finite inequality, including its normalization, and then pass from it to arbitrary orders. The method is the finite sampling form of flag algebras [7].

On vertices $0, \ldots, m-1$, list unordered pairs lexicographically. Encode a digraph by two bits for each pair $i < j$, the first recording $i \to j$ and the second $j \to i$. A five-vertex type is represented by the least mask under all vertex permutations. For such a type H, put

$$q(H) = \frac{1}{60} \sum_{v \in V(H)} d_H^+(v)(d_H^+(v) - 1).$$

Thus $q(H)$ is the probability that a uniform ordered triple of distinct vertices (v, x, y) has both $v \to x$ and $v \to y$.

We use three flag matrices. The first has one labelled root and two unlabelled additional vertices. The 64 labelled three-vertex digraphs give 36 flags after identifying the interchange of the two additional vertices. We order these flags by their least masks. For a five-vertex host H, choose the root in five ways and the first additional two-set in six ways; the complementary two-set gives the second flag. The resulting count matrix, divided by 30, is $M_1(H)$.

The other two matrices have three ordered roots and one additional vertex. The root masks are 31 and 63, respectively, a bidirected triangle with one arc removed and a bidirected triangle. For either root, the new vertex has $4^3 = 64$ possible adjacency patterns, ordered by the six-bit integer recording its two arcs with each root. For each of the 120 permutations of the host, use the first three vertices as roots and the last two as the two extensions. Count the flag pair if the ordered roots have the specified mask, and contribute zero otherwise. Division by 120 gives $M_{31}(H)$ and $M_{63}(H)$. These are unconditional matrices; no further division by the root density is made. All three matrices are symmetric.

For matrices of equal size, $\langle A, B\rangle = \sum_{i,j} A_{ij}B_{ij}$. The off-diagonal entries occur twice in this convention.

Lemma 2.1 (Rational certificate)**.** *There are rational positive semidefinite matrices Q_1, Q_{31}, Q_{63} of orders $36, 64, 64$ such that every five-vertex type $H \in \mathcal{C}$ satisfies*

$$q(H) + \sum_{b\in\{1,31,63\}} \langle Q_b, M_b(H)\rangle + s(H) = \frac{9}{16}, \qquad s(H) \geq 0. \tag{5}$$

The zero-slack types are exactly the complete bidirected multipartite graphs with part sizes

$$(5),\ (4,1),\ (3,2),\ (3,1,1),\ (2,2,1),\ (2,1,1,1).$$

Every other allowed type has

$$s(H) \geq \delta := \frac{52353}{2500000} > \frac{1}{50}. \tag{6}$$

Proof. The matrices are given by the exact rational certificate in the accompanying directory `supplement/tournaments`. We describe the finite verification so that its assertion is independent of numerical optimization.

The generator lists all 4^{10} labelled five-vertex digraphs. Independently, it obtains the twelve tournament classes by canonicalizing all 2^{10} labelled tournaments under the 120 permutations. A host is allowed if its arcs do not contain a labelled copy of at least one tournament class. The test permits extra reverse arcs, as required for ordinary subdigraph containment. There are 1 043 695 allowed labelled hosts and 9558 isomorphism types. The flag matrices are constructed by the placements just described. Their integer entries are stored in `flags.json`.

The certificate specifies integer matrices K_b and rational symmetric matrices R_b with

$$Q_b = K_b R_b K_b^{\mathsf{T}}.$$

The orders of R_1, R_{31}, R_{63} are $35, 64, 63$. After clearing a positive common denominator, fraction-free elimination verifies that every leading principal minor is strictly positive. Sylvester's criterion gives $R_b \succ 0$, and hence $Q_b \succeq 0$. The program then checks (5) over the rational numbers for all 9558 types. Exactly the six listed types have zero slack, and the least other slack is (6). Their masks, in increasing order, are

$$0,\ 255,\ 16380,\ 16383,\ 261375,\ 262143.$$

Two separate checks reconstruct the enumeration and flag products. The coverage check orients all digons in each of the 3^{10} semicomplete labelled hosts and records which tournaments occur. It finds 54168 allowed semicomplete hosts; adding the $4^{10} - 3^{10}$ hosts with a nonadjacent pair gives 1 043 695. The disjoint permutation orbits of the stored representatives have exactly this total size. The flag check uses all 120 placements for the one-root block, obtaining four times each stored integer entry, and recomputes both three-root blocks directly from their arcs. Thus the checks establish both the full set of hosts and every coefficient in the rational identity. The verification uses integer and rational arithmetic throughout; no semidefinite solver is required. □

To apply the finite identity at arbitrary orders, we need nonnegativity of the averaged flag terms. This comes from independent extensions of common roots; it is not asserted for each host separately. We remove the discrepancy between independent and disjoint sampling by taking blow-ups. The class $\mathcal{C}$ is closed under independent blow-ups, since a five-set meeting a fibre twice has a nonadjacent pair and contains no five-vertex tournament, while every other five-set inherits its type from the base digraph.

Lemma 2.2. *Let $D \in \mathcal{C}$ have $n \geq 1$ vertices. Choose five vertex labels independently and uniformly from $V(D)$, and place them on five distinct positions, declaring repeated labels nonadjacent. Let $p(D)$ be the probability that the resulting type is not one of the six zero-slack types. Then*

$$\frac{9}{16} - \frac{Q(D)}{n^3} \geq \delta p(D). \tag{7}$$

In particular, $Q(D) \leq 9n^3/16$.

Proof. Replace every vertex of D by t independent clones to obtain $D(t)$. It belongs to $\mathcal{C}$ and satisfies $Q(D(t)) = t^3 Q(D)$. Average (5) over uniform induced five-sets in $D(t)$.

Conditional on an ordered root, choose the two extensions independently. Each flag product matrix then has expectation ff^{T}, where f is the conditional flag probability vector; it is zero if the root has the wrong type. Its inner product with the fixed positive semidefinite Q_b is nonnegative. Requiring the two extensions to be disjoint changes each expectation by $O(1/(tn))$, since the number of additional vertices is bounded. This estimate also applies to the one-root block, whose extensions are two-sets.

The average of $q(H)$ is the normalized count of ordered distinct vertices supporting two out-arcs. Writing $d^+(v) = d_D^+(v)$, it equals

$$\frac{\sum_{v \in V(D)} t\,(td^+(v))(td^+(v) - 1)}{tn(tn-1)(tn-2)} \longrightarrow \frac{Q(D)}{n^3}.$$

The induced host distribution converges to the independent-label experiment in the statement. Averaging the nonnegative slacks, using (6), and letting $t \to \infty$ proves (7). □

The homogeneous equality case follows already from this gap. If $Q(D) = 9n^3/16$, then $p(D) = 0$. A single-arc pair would occur in the first two positions with positive probability and would force a positive-slack type. Thus every adjacent pair is a digon. An induced triple consisting of one digon and an isolated vertex is equally impossible. Consequently nonadjacency, together with equality, is an equivalence relation: a failure of transitivity would give exactly that triple. Therefore D is complete bidirected multipartite. It has at most four nonempty parts, since a bidirected K_5 violates $\mathcal{C}$.

For part proportions $x_1, \ldots, x_r$, the normalized objective is $\sum_i x_i(1 - x_i)^2$. Pairwise averaging gives

$$f(a) + f(b) - 2f((a+b)/2) = -\frac{(a-b)^2(4 - 3a - 3b)}{4}, \qquad f(x) = x(1-x)^2. \tag{8}$$

For $a + b \leq 1$, the coefficient is positive. Hence the unique maximum for fixed r is $(1 - 1/r)^2$, attained by equal parts. For $r \leq 4$, equality with $9/16$ requires four equal parts. Conversely, that graph attains the bound.

To obtain the finite exact extremum, we need a reduction for exact maximizers. The cloning method appears in [4, 3]; we include the details needed here.

Lemma 2.3. *Fix a tournament F. Every n-vertex F-free maximizer of Q can be transformed into an independent blow-up of a semicomplete digraph while preserving the independent classes of nonadjacency and the out-degree of each labelled vertex.*

Proof. For nonadjacent u, v, let $D_{v\leftarrow u}$ replace v by an independent clone of u. This preserves F-freeness: a copy of a tournament cannot use both clones, and a copy using only the new clone can replace it by u. Direct expansion gives

$$Q(D_{v\leftarrow u}) + Q(D_{u\leftarrow v}) - 2Q(D) = 2 \sum_{w\notin\{u,v\}} (\mathbf{1}_{w\to u} - \mathbf{1}_{w\to v})^2.$$

Both differences on the left are nonpositive by maximality. Thus the in-neighborhoods of u, v agree. The change under a single cloning is now $d^+(u)^2 - d^+(v)^2$, so the two out-degrees agree as well. Each cloning preserves all labelled out-degrees.

If u is nonadjacent to both v, w, equality of in-neighborhoods implies $v \not\to w$ and $w \not\to v$. Nonadjacency is therefore an equivalence relation. Its classes are independent. Clone a representative within each class. For a vertex outside that class, either it points to every class member, or every class member points to it; this ensures that cloning preserves the nonadjacency classes. Processing the classes in succession also preserves any adjacency rows already made equal. The result has identical in- and out-neighborhoods within each class, so its quotient is semicomplete. All claimed data have been preserved. □

Finite exact part of Theorem 1.1. Let F be any five-vertex tournament, and let D be an n-vertex F-free maximizer. Let D^* be the blow-up provided by Lemma 2.3. It has the same objective. If A is the number of its single-arc pairs, then the independent-label experiment gives $p(D^*) \geq 2A/n^2$. Equations (7) and (6), together with $Q(D^*) \geq B_4(n)$, imply

$$\frac{2An}{50} \leq \frac{9n^3}{16} - Q(D^*) \leq \frac{9n^3}{16} - B_4(n) \leq \frac{5n}{4}.$$

For the last inequality, the four residues of n modulo four give respectively 0, $(15n-6)/16$, $5n/4$, and $(15n+6)/16$. Therefore $A \leq 31$.

Let S consist of the endpoints of all single-arc pairs, and set $s = |S| \leq 62$. If two independent classes of D^* have single arcs between them, every vertex in both classes belongs to S. Hence S is a union of whole classes. The remaining digraph is complete bidirected multipartite with at most four parts.

Suppose $A > 0$. Every pair joining S to its complement is a digon: its vertices lie in different classes, so they are adjacent, and a single arc would put both endpoints in S. If the complement had four nonempty parts, one vertex of S and one from each of those parts would form a bidirected K_5, which contains F. Thus the complement has at most three parts. Let their sizes be h_1, h_2, h_3, allowing zero parts, with sum $n - s$. Pairwise averaging, as in (8), gives

$$\begin{aligned} Q(D^*) &\leq \sum_{i=1}^{3} h_i(n-h_i)^2 + sn^2 \leq (n-s)\left(n - \frac{n-s}{3}\right)^2 + sn^2 \\ &= \frac{4}{9}n^3 + sn^2 - \frac{ns^2}{3} - \frac{s^3}{9} \leq \frac{4}{9}n^3 + 62n^2. \end{aligned}$$

For $n \geq 526$, this is smaller than $9n^3/16 - 5n/4 \leq B_4(n)$. The comparison is equivalent to $17n^2 - 8928n - 180 > 0$, which holds at 526 and is increasing thereafter. This contradiction proves $A = 0$.

Now D^* is complete bidirected multipartite with at most four parts. Splitting a part of size a into sizes $b, a-b$ increases Q by $b(a-b)(4n-3a) > 0$. Moving a vertex from a part of size a to one of size b, with $a \geq b+2$, increases it by $(a-b-1)(4n-3a-3b) > 0$. Thus the unique optimal partition has four parts of sizes differing by at most one. The original digraph has these same independent classes and the same labelled out-degrees. Every vertex consequently points to every vertex outside its class, so the original graph is also $\overleftrightarrow{T}_4(n)$. □

3 Stability of the original digraph

Cloning an extremizer does not by itself control the distance of a near-extremizer from a multipartite graph. We instead apply the gap directly to the original digraph and repair its digon graph. The following elementary lemma supplies the required quantitative repair.

Lemma 3.1. *If an n-vertex undirected graph has T induced copies of the three-vertex path, then at most $\sqrt{6Tn}$ edge additions and deletions turn it into a disjoint union of cliques.*

Proof. When $T = 0$, adjacency together with equality is an equivalence relation. Assume $T > 0$ and put $\alpha = \sqrt{3T/(2n^3)}$. In the current induced graph J, if $e(J) \geq \alpha n^2$, consider the closed neighborhood $C_v = N_J[v]$ of a vertex v. The cost E_v of completing C_v to a clique and deleting all edges between C_v and its complement is exactly the number of induced three-vertex paths containing v. Missing edges inside C_v correspond to paths with middle vertex v; edges from C_v to the outside correspond to paths with endpoint v. Hence

$$\sum_v E_v = 3T(J) \leq 3T, \qquad \sum_v |C_v| = |V(J)| + 2e(J) \geq 2\alpha n^2.$$

Choose v with $E_v/|C_v| \leq 3T/(2\alpha n^2)$, perform the edits, and remove C_v. The remaining graph is still an induced subgraph of the original graph. The selected sets are disjoint, so the total cost of these steps is at most $3T/(2\alpha n)$. When fewer than αn^2 edges remain, delete them all. The total is at most $3T/(2\alpha n) + \alpha n^2 = \sqrt{6Tn}$. □

Stability part of Theorem 1.1. In fact, the argument applies to every $D \in \mathcal{C}$. Put $\rho = \varepsilon/\delta$, where δ is from (6). The gap gives $p(D) \leq \rho$. If A is the number of single-arc pairs, then

$$A \leq \frac{\rho n^2}{2}, \qquad T \leq \frac{\rho n^3}{6},$$

where T counts triples spanning exactly one edge in the undirected digon graph G. The first assertion follows by placing a single-arc pair in the first two sampled positions. For the second, a triple counted by T either contains a single-arc pair in D, or consists of one digon and an isolated vertex. Both patterns force positive slack regardless of the last two sampled labels. The probability of this ordered triple event is exactly $6T/n^3$. Labels in the last positions may repeat, so no finite-order error is needed.

The graph G is K_5-free. Its complement has exactly T induced three-vertex paths. Apply Lemma 3.1 to obtain a complete multipartite graph B within κn^2 undirected edits of G, where $\kappa = \sqrt{\rho}$. After deleting the single arcs of D and changing each edited pair in both directions, the distance from D to the bidirected version of B is at most $(\rho/2 + 2\kappa)n^2$.

Let the part proportions of B be $x_1 \geq x_2 \geq \cdots$, padded with zeros, and write $s = 1 - x_1 - x_2 - x_3 - x_4$. In an independent five-vertex sample, after each of the first four choices the mass outside the chosen parts is at least s. Thus the ordered K_5 density of B is at least s^4. Every such clique

uses an edge of B absent from G. Each of the at most κn^2 added edges lies in at most $20n^3$ ordered five-tuples. Consequently

$$s^4 \le 20\kappa. \tag{9}$$

Merging every part after the fourth into the first changes at most $2sn^2$ arcs. The resulting bidirected graph B_4 has at most four parts, and its distance from D is at most $\eta_0 n^2$, where

$$\eta_0 = \rho/2 + 2\sqrt{\rho} + 2(20\sqrt{\rho})^{1/4}.$$

Changing one arc changes Q by at most $2n$. If the four proportions of B_4 are $y_i = 1/4 + z_i$, then $\sum_i z_i = 0$ and

$$\frac{Q(B_4)}{n^3} = \frac{9}{16} - \frac{5}{4}\sum_i z_i^2 + \sum_i z_i^3 \le \frac{9}{16} - \frac{1}{2}\sum_i z_i^2,$$

since $z_i \le 3/4$. Comparison with $Q(D)$ gives

$$\sum_i z_i^2 \le 2(\varepsilon + 2\eta_0). \tag{10}$$

Equations (9) and (10) show that all four largest original parts of B have mass at least $1/8$ when ε is sufficiently small. This preliminary conclusion improves the tail bound. Choosing one vertex in each of those four parts and one in the tail gives

$$\frac{120}{4096}s \le 120x_1x_2x_3x_4s \le 20\kappa, \qquad s \le \frac{2048}{3}\kappa.$$

The same merging now yields a four-part graph within $\eta_1 n^2$ arcs of D, where

$$\eta_1 = \rho/2 + \frac{4102}{3}\sqrt{\rho}.$$

Repeating the square-sum comparison gives $\sum_i z_i^2 \le 2(\varepsilon + 2\eta_1)$, and hence $\sum_i |z_i| \le 2\sqrt{2(\varepsilon + 2\eta_1)}$.

The four integer part sizes can be balanced by moving at most $(n/2)\sum_i |z_i| + 3/2$ vertices. Each move costs at most $2n$ arc edits. We therefore obtain total distance at most

$$(\eta_1 + 2\sqrt{2(\varepsilon + 2\eta_1)})n^2 + 3n = O(\varepsilon^{1/4})n^2 + 3n.$$

This proves the assertion for small positive ε. The zero case was established after Lemma 2.2. For larger ε, enlarge the absolute constant, using the universal distance bound $n(n-1)$. This proves the theorem for all $\varepsilon \ge 0$. □

4 Bidirected joins and convex moments

The preceding tournament result does not extend to every forbidden digraph with the same ordinary extremal function. We prove Theorem 1.2 by establishing its ordinary extremum, constructing a graph that attains every prefix degree bound, and recovering its structure from equality. The same prefix bounds then give a stability theorem for square-sum near-extremizers.

We use the following weighted form of the ordinary tournament argument, due to Norin [6]. A set in a graph with pair weights in $\{0, 1, 2\}$ is an *anti-matching* if all its pair weights are positive and its weight-one pairs form a matching.

Lemma 4.1 (Norin)**.** *If a weighted graph on n vertices contains no anti-matching of size $r \ge 2$, then its total pair weight is at most $2t_{r-1}(n)$.*

Proof. We include the induction and its integer comparison. The assertion is immediate for $n < r$. Choose a nonempty anti-matching S of maximum internal weight and let $s = |S| \le r-1$. Every outside vertex has total weight at most $2s-2$ to S. A larger weight is either $2s$, allowing the vertex to be added, or $2s-1$, with a unique weight-one pair to some $u \in S$. If u has no weight-one neighbor in S, add the new vertex. Otherwise replace u by it; this increases internal weight by one. Each alternative contradicts the choice of S.

Deleting S therefore removes weight at most $(s-1)(2n-s)$. Put $m = r-1$. The integer Turán difference $t_m(j) - t_m(j-1) = j - \lceil j/m \rceil$ gives

$$2t_m(n) - 2t_m(n-s) - (s-1)(2n-s) = 2n - 2\sum_{j=0}^{s-1} \left\lceil \frac{n-j}{m} \right\rceil \ge 0.$$

Indeed, the sum over m consecutive terms is n, and all those terms are nonnegative; here $s \le m$ and $s \le n$. Apply induction to the graph outside S and add the deletion bound to obtain $2t_m(n)$. □

Proof of (1). Give every unordered pair in a digraph weight equal to its number of arcs. By Lemma 4.1, more than $2t_{2q}(n)$ arcs force a $(2q+1)$-vertex anti-matching. Complete its single-arc pairs to a matching of size q, leaving one vertex unpaired. One matched pair and the unpaired vertex support C_3: both pairs incident with the unpaired vertex are digons, so either orientation of the matched pair can be completed to a directed triangle. The other matched pairs support the $q-1$ copies of T_2, and every cross-block pair is a digon. Thus this anti-matching contains F_q.

Conversely, $\overleftrightarrow{T}_{2q}(n)$ is F_q-free because its underlying graph has no K_{2q+1}, whereas the underlying graph of F_q is complete. It has $2t_{2q}(n)$ arcs, proving the equality. □

The ordinary extremum bounds every prefix of a decreasing out-degree sequence. To use the convex-moment theorem of [1, Theorem 2.2], one graph must attain all these bounds simultaneously. Place vertices $0, \dots, n-1$ into q classes by the value of $\lfloor j/2 \rfloor$ modulo q. Within each class, consecutive assigned pairs form digon blocks, and earlier blocks send all arcs to later ones. All cross-class pairs are digons. Denote the result by $D_{q,n}$.

The graph $D_{q,n}$ is F_q-free. A copy of F_q uses at most two vertices in any class. If they come from one forbidden block, this follows from the absence of a directed triangle in a chain and the size of T_2. If at least three vertices in one class came from two or more forbidden blocks, one would require two digon neighbors within that class, but the digons of a chain form a matching. The q classes therefore cannot contain the $2q+1$ vertices of F_q.

A vertex j misses precisely the two vertices of each earlier block in its own class, and hence

$$d^+_{D_{q,n}}(j) = b_j := n - 1 - 2\left\lfloor \frac{j}{2q} \right\rfloor. \tag{11}$$

This is a nonincreasing sequence. For the prefix $U_k = \{0, \dots, k-1\}$, write $k = 2qa + b$, $0 \le b < 2q$. The class sizes s_i in this prefix lie in $\{2a, 2a+1, 2a+2\}$. The $2q$ integers $\lceil s_i/2 \rceil, \lfloor s_i/2 \rfloor$ are balanced, so

$$e(D_{q,n}[U_k]) = k^2 - \sum_i s_i^2 + \sum_i \lfloor s_i^2/2 \rfloor = 2t_{2q}(k).$$

All arcs from U_k to its complement are present, including when the cut splits a digon block. Therefore

$$\sum_{j<k} b_j = 2t_{2q}(k) + k(n-k). \tag{12}$$

Proof of (2). List the out-degrees of any F_q-free digraph as $a_0 \geq \cdots \geq a_{n-1}$. Applying (1) to the vertices of its first k entries and using (12) gives

$$\sum_{j<k} a_j \leq 2t_{2q}(k) + k(n-k) = \sum_{j<k} b_j \qquad (1 \leq k \leq n). \tag{13}$$

This is the prefix-majorization method of [1, Lemma 2.1 and Theorem 2.2]. For completeness, each nonnegative integer threshold t satisfies

$$\sum_j (a_j - t)_+ = \max_{0\leq k\leq n} \left(\sum_{j<k} a_j - kt \right) \leq \max_{0\leq k\leq n} \left(\sum_{j<k} b_j - kt \right) = \sum_j (b_j - t)_+.$$

On the integer interval $0, \ldots, n-1$, a nondecreasing convex function has the representation

$$\varphi(x) = \varphi(0) + c_0 x + \sum_{t=1}^{n-2} c_t (x-t)_+, \quad c_0 = \varphi(1) - \varphi(0) \geq 0, \quad c_t = \varphi(t+1) - 2\varphi(t) + \varphi(t-1) \geq 0.$$

For $n = 1$, the assertion is immediate and this representation is unnecessary. Summing the representation and the threshold inequalities proves the upper bound. The graph $D_{q,n}$ attains it by (11). Since F_q is self-converse, reversing arcs gives the identical in-degree extremum. □

The majorization argument determines the extremal value. To classify the extremizers, we first recover their degree sequence and then use equality in each prefix bound to recover the arcs. If equality holds for the square function, then $a_j = b_j$ for every j. Indeed,

$$x^2 = x + 2 \sum_{t=1}^{n-2} (x-t)_+$$

has positive coefficients, so equality forces equality of the total sum and of every positive-part sum. Differences of consecutive positive-part sums determine the multiplicities of every positive integer degree, and the number of vertices determines the multiplicity of zero. For a strictly convex nondecreasing function, all c_t are positive, and $c_0 > 0$ when $n \geq 2$. The same argument applies.

Extremal classification in Theorem 1.2. Let D attain the square extremum and order its vertices by nonincreasing degree. Since the degree sequence is exactly (11), equality holds in both terms of every prefix bound (13). All arcs from an earlier vertex to a later vertex are therefore present. Vertices of equal degree may be reordered arbitrarily, so every degree layer is complete bidirected.

Write $n = 2qa + b$, $0 \leq b < 2q$, as in the statement. The full layers $W_0, \ldots, W_{a-1}$ each have $2q$ vertices, with degree $n - 1 - 2t$ on W_t; a final layer W_a has size b if $b > 0$. For $v \in W_t$ and each earlier full layer W_s, at least two arcs from v to W_s must be absent. Otherwise $W_s \cup \{v\}$ is an anti-matching and contains F_q. The total degree deficit of v is exactly $2t$, while all same-layer and forward arcs are present. Thus exactly two outgoing arcs are missing into each earlier layer.

If $n \leq 2q$, every extremizer is the complete bidirected graph, as specified by (3). Assume $n > 2q$, put $S = W_0$, and for $v \notin S$ define

$$A(v) = \{x \in S : v \not\to x\}.$$

Each $A(v)$ is a pair. Two observations determine the classes.

First, if $u, v \notin S$ form a digon, their pairs $A(u), A(v)$ are either equal or disjoint. If they met in exactly one vertex x, the set $(S \setminus \{x\}) \cup \{u, v\}$ would be an anti-matching: its only possible single-arc pairs are disjoint. It would contain F_q, a contradiction.

Second, for a pair $P \subseteq S$, the set $X_P = \{v \notin S : A(v) = P\}$ induces a C_3-free graph. A directed triangle in it, together with a pairing of $S \setminus P$, would give F_q, because all the cross arcs are present. In particular, at most two vertices of any one degree layer can have the same A-pair.

If $2q < n \leq 4q$, there is only one nonempty layer outside S. Its vertices are pairwise bidirected, so their distinct A-pairs are disjoint and each is used at most twice. Complete these pairs to a perfect matching $P_1, \ldots, P_q$ of S. Each P_i together with the outside vertices using it is an ordered digon chain of size two, three, or four; all pairs between different chains are digons. This is exactly (3) in this range, including $n = 4q$.

Now let $n > 4q$. The layer W_1 has size $2q$, so its A-pairs form a perfect matching $P_1, \ldots, P_q$ of S, each used twice. If a later vertex v had an A-pair other than these P_i, it would meet two of them once. The first observation would prohibit a digon between v and all four corresponding vertices of W_1. All forward arcs are present, so v would miss at least four arcs into W_1, contradicting its exact deficit of two. Thus every outside vertex uses one of the fixed pairs P_i.

Every full layer uses each P_i exactly twice, by the second observation and its size $2q$. The last layer uses each pair zero, one, or two times. Let $V_i = P_i \cup X_{P_i}$. A later vertex in V_i has no outgoing arc to either of the two vertices of an earlier layer in V_i: for the initial layer this is the definition of $A(v)$, and for any other layer a reverse arc would form a directed triangle in X_{P_i}. Since these are exactly the two missing arcs into that layer, all its arcs to the other classes are present. Each V_i is therefore an ordered digon chain, and all cross-class pairs are digons. Its size is $2a + s_i$ with $s_i \in \{0, 1, 2\}$ and $\sum_i s_i = b$, proving necessity.

Conversely, each graph in (3) is F_q-free by the same two-vertices-per-class argument used for $D_{q,n}$. Its vertices in layer t have degree $n - 1 - 2t$, so its degree sequence is b. It attains every convex upper bound. The strict-convex equality argument above proves the claimed extended classification, and reversal proves the in-degree statement. □

The asymptotic formula (4) is the Riemann sum of $\int_0^1 (1 - x/q)^2 \, dx$. There are also small exact witnesses to the failure of the proposed tournament-type bound. For $n = 4q$, the graph $D_{q,n}$ has

$$e(D_{q,n}) = 16q^2 - 8q, \qquad Q(D_{q,n}) = 64q^3 - 64q^2 + 20q.$$

Its excess over $((2q-1)/(2q))^2(4q)^3$ is $4q > 0$. In particular, $q = 2$ gives eight vertices, 48 arcs, and square sum $296 > 288$. Formula (2) shows that the maximum square sums for this forbidden graph at orders one through seven are $0, 2, 12, 36, 68, 118, 192$, respectively, so this eight-vertex witness is smallest for F_2.

For near-extremizers, the same prefix deficits control the number of missing forward arcs in degree order. The digon graph need not itself be K_{q+1}-free. Instead, almost all forward arcs being present forces it to contain few copies of the complete $(q+1)$-partite graph with two vertices in each part. Graph removal and stability then recover the partition, and the degree order supplies the chains inside its parts.

Theorem 4.2. *Fix $q \geq 2$. For every $\eta > 0$ there are $\delta > 0$ and N such that every n-vertex F_q-free digraph D with $n \geq N$ and*

$$Q(D) \geq \mathrm{ex}_2^+(n, F_q) - \delta n^3$$

is at distance at most ηn^2 from a bidirected join of q ordered digon chains whose sizes differ by at most one. In particular, the target is an exact square-sum extremizer.

Proof. It suffices to prove the sequence formulation. Fix q and suppose $\Delta = \mathrm{ex}_2^+(n, F_q) - Q(D) = o(n^3)$. Order the vertices by nonincreasing out-degree $a_0, \ldots, a_{n-1}$, and let b_j be the sequence

in (11). Put $S_0 = 0$ and $S_k = \sum_{j<k}(b_j - a_j)$. By (13), all S_k are nonnegative. Summation by parts gives

$$\sum_{j=0}^{n-1} a_j(b_j - a_j) = a_{n-1}S_n + \sum_{k=1}^{n-1}(a_{k-1} - a_k)S_k \geq 0.$$

Consequently,

$$\sum_j (b_j - a_j)^2 \leq \Delta, \qquad 0 \leq S_k \leq \sqrt{n\Delta} \quad (1 \leq k \leq n), \tag{14}$$

since $\Delta = \sum_j (b_j - a_j)^2 + 2\sum_j a_j(b_j - a_j)$ and Cauchy–Schwarz bounds each prefix sum.

For $U_k = \{0, \ldots, k-1\}$, let ℓ_k count the missing arcs from U_k to its complement. The exact decomposition

$$S_k = 2t_{2q}(k) - e(D[U_k]) + \ell_k$$

implies $\ell_k \leq S_k$, because $D[U_k]$ is F_q-free. A missing forward arc $i \to j$, with $i < j$, is counted in precisely $j - i$ of the numbers ℓ_k. Thus, if m is the total number of missing forward arcs and $\varepsilon = \Delta/n^3$, then for every positive integer r,

$$m \leq rn + \frac{1}{r}\sum_{k=1}^{n-1} \ell_k \leq rn + \frac{\sqrt{\varepsilon}\, n^3}{r}.$$

Here fewer than rn pairs have $j - i < r$, and each remaining missing arc contributes at least r to the sum. Taking $r = \max\{1, \lceil \varepsilon^{1/4} n \rceil\}$ yields

$$m \leq 2\varepsilon^{1/4} n^2 + n = o(n^2). \tag{15}$$

Let G be the undirected graph whose edges are the digons of D. Each pair contributes at most one arc in addition to its digon indicator, so (14) with $k = n$ gives

$$\begin{aligned} e(G) &\geq e(D) - \binom{n}{2} \geq 2t_{2q}(n) - \binom{n}{2} - \sqrt{n\Delta} \\ &= \frac{1}{2}\left(1 - \frac{1}{q}\right) n^2 - o(n^2). \end{aligned}$$

Let $H_q = K_{q+1}(2)$ be the complete $(q+1)$-partite graph with two vertices in each part, and put $h = 2q + 2$. Every copy of H_q in G contains a pair missing its forward arc in D. Indeed, suppose all forward arcs on its vertices were present. Take its earliest vertex x, in a part A, and the two vertices $y < z$ of another part B. The arcs $x \to y$, $y \to z$, and $z \to x$ form a directed triangle. The remaining $q - 1$ parts each supply a forward arc, and all pairs between these forbidden blocks are digons. These vertices therefore contain F_q, a contradiction.

A fixed missing forward pair lies in at most $h(h-1)n^{h-2}$ labelled copies of H_q. Equation (15) implies that G has $o(n^h)$ such copies. The graph removal lemma [8, Theorem 2, with $k = 2$] deletes $o(n^2)$ edges to produce an H_q-free graph G_0. It still has at least $(1 - 1/q)n^2/2 - o(n^2)$ edges. Since $\chi(H_q) = q + 1$, the Erdős–Simonovits stability theorem, in the form stated in [2, Section 1], gives a balanced complete q-partite graph T with

$$|E(G) \triangle E(T)| = o(n^2).$$

To recover the directed structure, add the $m = o(n^2)$ missing forward arcs to D. Its digon graph changes in at most m pairs. With every forward arc present, the reverse arcs are precisely the edges of this digon graph. Changing their set to $E(T)$ therefore costs another $o(n^2)$ arc edits. The

resulting digraph has all cross-part digons and a transitive tournament inside each part, ordered as above. Pair successive vertices within each part and add the reverse arc on every pair. At most $n/2$ further edits turn the parts into ordered digon chains. If $n = 2qa + b$ and $0 \le b < 2q$, their balanced sizes are $2a + s_i$ with $s_i \in \{0, 1, 2\}$ and $\sum_i s_i = b$. By Theorem 1.2, this is an exact extremizer. The total distance is $o(n^2)$, proving the assertion. □

Theorem 4.2 gives a qualitative conclusion for each fixed q; it does not provide an explicit edit bound or stability for every convex objective. Theorems 1.2 and 4.2 concern only the family F_q. Theorem 1.1 leaves the remaining small-order cases for nonregular five-vertex tournaments open.

Data availability and computational methods. The exact certificate, verification programs, and instructions are included in the accompanying `supplement` directory. Its entry point is `run_all.py`. GPT-6.1 (OpenAI) assisted with implementing the programs. Section 2 specifies the exact arithmetic and coverage checks on which the computer-assisted proof relies; the finite checks for chain joins supplement the general proof in Section 4.

Declaration of AI use. The author used GPT-6.1 to assist with exploring constructions, checking calculations, preparing verification programs, and improving the exposition. The author takes full responsibility for the content of this paper.

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